

ITMA-8891-ROINUJ

Answer All Questions :

Time : $1\frac{1}{2}$ Hours

1. The number $5\sqrt{7}$ lies between
 (a) 10 and 11 (b) 11 and 12 (c) 12 and 13 (d) 13 and 14
2. The eliminant of x and y from the equations
 $x+y = a$, $x-y = b$, $x^2+y^2-4xy = c$ is
 (a) $a^2-3b^2-2c = 0$ (b) $3a^2-b^2-2c = 0$
 (c) $a^2-3b^2+2c = 0$ (d) $a+b=4c$.
3. The sum of the series $1^2-2^2+3^2-4^2+5^2-6^2+\dots$ to 99 terms is
 (a) -4950 (b) 5050 (c) 163350 (d) 10000
4. The number of numbers from 1 to 99 containing neither 1 nor 2 is
 (a) 62 (b) 64 (c) 60 (d) 80
5. If a, b, c are in Arithmetic progression, then one of the roots of $ax^2+2bx+c=0$ is
 (a) 0 (b) 1 (c) $-b/a$ (d) $-c/a$
6. If $f(n)$ denotes the largest prime not larger than n , for each natural number greater than 1, then
 $f(2) + f(3) + f(4) + \dots + f(10)$ is
 (a) 54 (b) 50 (c) 46 (d) 47
7. If $x^2 = x+1$ implies $x^5 = ax+b$, then the product ab is
 (a) 6 (b) 2 (c) 15 (d) 40
8. If $a*b = a+b-ab$ and $a_0b = a^b$, then
 $(3*4)_02 - 3*(4_02)$ is equal to
 (a) 0 (b) -2 (c) 52 (d) 2

9. Three identical coins each 3 cm. in diameter are placed on a table touching one another. An elastic band is passed around them and held taut. The length of the stretched band in cm. is
 (a) $3\pi+9$ (b) $6\pi+9$ (c) $3\pi+18$ (d) $6\pi+18$.
10. The product of four consecutive natural numbers is always divisible by
 (a) 9 (b) 4 but not 8 (c) 8 or a higher power of 2 (d) 16.
11. The number of circles passing through three given points is
 (a) 0 or 1 (b) 1 or ∞ (c) 2 or ∞ (d) always 1.
12. The speed of a boat up a stream and down it are a km/h and b km/h respectively. Then the average velocity in km/h is
 (a) $\frac{a+b}{2}$ (b) $\sqrt{ab}$
 (c) $\frac{2ab}{a+b}$ (d) $2|a-b|$.
13. The number to be subtracted to each of the numbers 2, 4, 7 so that the resulting numbers are in Geometric Progression is
 (a) -1 (b) 1 (c) -2 (d) 2.
14. Let a and b be positive rationals. If $\sqrt{ab}$ is rational, then so also is
 (a) $\sqrt{a+b}$ (b) $\sqrt{|a-b|}$ (c) $\sqrt{\frac{b}{a}}$ (d) $\sqrt{a} + \sqrt{b}$
15. The equation $x^2 + |x| = 6$ has the solutions
 (a) $x = 2, x = 3$ (b) $x = 3, x = -2$
 (c) $x = \pm 3$ (d) $x = \pm 2$.
16. ABC is a triangle right angled at A . If $AC = 3, BC = 4$, then the length of the altitude through C is
 (a) 2.4 (b) $\frac{12}{7}$ (c) 4.8 (d) 1.2.

17. The least value of $3^x + 3^{-x}$ is
 (a) 1 (b) $\frac{1}{3}$ (c) 2 (d) $\frac{2}{3}$.
18. If $(1 + x + 2x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, then $a_0 - a_1 - a_2 - a_3 + \dots + a_{2n}$ is equal to
 (a) 0 (b) 2^n (c) 4^n (d) $n!$.
19. If A and B are two sets, then the negation of $x \in A - B$ is equivalent to
 (a) $x \notin A$ and $x \in B$ (b) $x \notin A$ and $x \notin B$
 (c) $x \notin A$ or $x \in B$ (d) $x \in A$ or $x \notin B$.
20. If an article bought for Rs. 8,000/- is worth Rs. 4500/- at the end of two years then its value at the end of three years to the nearest Rupee (the annual rate of depreciation being constant) is
 (a) Rs. 2750 (b) Rs. 3375 (c) Rs. 2531 (d) Rs. 2532
21. For any finite set A , let $n(A)$ denote the number of elements in A . The number of elements in the cartesian product $A \times B$ with distinct co-ordinates is
 (a) $n(A) \cdot n(B)$ (b) $n(A) \cdot n(B) - n(A \cup B)$
 (c) $n(A) \cdot n(B) - n(A \cap B)$ (d) $n(A) + n(B) - n(A \cap B)$
22. A rearrangement of the numbers $2^{\frac{1}{8}}$, $3^{\frac{1}{8}}$, $5^{\frac{1}{7}}$ in the increasing order is obtained by
 (a) the same as given
 (b) interchanging the second and third numbers
 (c) moving the first to the last position
 (d) interchanging the first and the third.
23. The ratio of the areas of an equilateral triangle and a square standing on the same base is
 (a) $3 : 4\sqrt{3}$ (b) $\sqrt{3} : 2$ (c) $3 : 16$ (d) $\sqrt{3} : 3$

24. The number $\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}}$ is equal to
 (a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) $2\sqrt{3}$ (d) $\sqrt{2}$.
25. The number of ways of choosing two non-adjacent vertices of a convex hexagon is
 (a) 6 (b) 8 (c) 9 (d) 15.
26. A square and a regular hexagon have the same area. The ratio of their sides is
 (a) $3 : 12^{\frac{1}{4}}$ (b) $3\sqrt{3} : 2$ (c) $3 : 2$ (d) $\sqrt{108} : 2$.
27. If $(a+b+c)^2 + (b+c-a)^2 + (c+a-b)^2 + (a+b-c)^2 = 4(bc+ca+ab)$, where a, b, c are real, then
 (a) $a+b+c = 0$ (b) $abc = 0$
 (c) $a=b=c = 0$ (d) $a^3+b^3+c^3 = 3abc$.
28. The line segment joining the points $(3,0)$ and $(0,1)$ in the xy -plane is rotated about the x -axis and the y -axis. The ratio of the volumes of cones so generated is
 (a) $1 : 9$ (b) $9 : 1$ (c) $3 : 1$ (d) $1 : 3$.
29. If $x = a$, $y = b$ is a solution of $5x+9y = 1$, then the value of y which corresponds to $x = a+9$ is
 (a) $b+9$ (b) $b-9$ (c) $b+5$ (d) $b-5$.
30. Two adjacent sides of a parallelogram are 31 and 17. If one diagonal is of length 40, the other diagonal is of length
 (a) 8 (b) 20 (c) 25 (d) 30.
31. A piece of work can be done by 4 men in 4 days working 4 hours a day. How many men will do the same piece of work in 2 days working 2 hours a day?
 (a) 2 (b) 8 (c) 16 (d) 32.
32. If $A = \{\phi, 1\}$ and $P(A)$ is the set of all subsets of A , which of the following statements is true?
 [ϕ denotes the empty set.]

(i) $\emptyset \in P(A)$ (ii) $\phi \subset P(A)$ (iii) $\{\phi\} \in P(A)$ (iv) $\{\phi\} \subset P(A)$

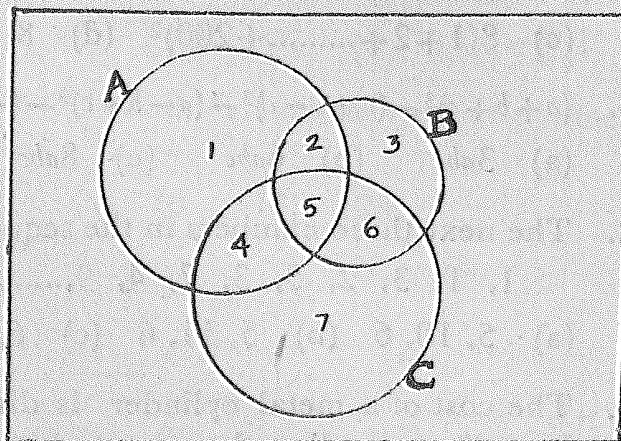
(a) (i) only

(b) (i) and (iii) only

(c) (i), (ii), (iii) only

(d) all the four statements (i) through (iv).

33. $(A \cup B) - (A \cap C)$
is the union
of regions
numbered



(a) 1, 2, 3, 4, 6

(b) 1, 2, 3

(c) 1, 2, 3, 5

(d) 1, 2, 3, 6

34. The product of 3 natural numbers (greater than 1) is 60. A possible sum of the three numbers is

(a) 16

(b) 11

(c) 18

(d) 19.

35. If $a+b = x$, $a^2+b^2 = y$, $a^3+b^3 = z$, then

(a) $2z = 3xy - x^3$ (b) $xy = z$ (c) $x\sqrt{y} = \sqrt[3]{z^2}$ (d) $x^6 + y^3 = 2z^2$.

36. The number of eggs in a basket doubles every minute. The basket is full of eggs in an hour. When was the basket half full?

(a) at the end of half an hour (b) at the end of 31st minute

(c) at the end of 59th minute (d) at the end of 2nd minute.

37. A painter painted the walls of a room in 3 days. The time needed to paint a room twice as long, wide and high is

(a) 6 days

(b) 12 days

(c) 24 days

(d) 48 days.

38. If 4, 6, a are in A.P., 4, 6, b are in G.P. and 4, 6, c are in G.P., then $a+b-c$ is

(a) 5

(b) 6

(c) 7

(d) 12.2.

39. A square and a circle have the same area. The ratio of the squares of their perimetres is
 (a) $2 : \sqrt{\pi}$ (b) $4 : \pi$ (c) $4 : \pi^2$ (d) $2 : \pi$.
40. $2^3 + 4^3 + 6^3 + 8^3 + \dots + 200^3$ is equal to
 (a) $8(1+2+\dots+100)^2$ (b) $8(1+2+\dots+100)^3$
 (c) $8(1+2+\dots+200)^2$ (d) $8(1+2+\dots+200)^3$.
41. $(a+b+c)^3 - (a+b-c)^3 - (a-b+c)^3 - (-a+b+c)^3$ is equal to
 (a) $3abc$ (b) $6abc$ (c) $8abc$ (d) $24abc$.
42. The next three numbers in the sequence
 1, 1, 3, 2, 5, 3, 7, 4, 9, is
 (a) 5, 10, 6 (b) 5, 11, 6 (c) 6, 10, 7 (d) 6, 11, 8.
43. The cost of a metal cylinder is directly proportional to its radius as well as the square of its length. If the cost of a cylinder with radius 2 cm. and length 6 cm. is Rs. 72.00. What will be its cost if the radius is halved without changing its volume?
 (a) Rs. 72.00 (b) Rs. 144.00
 (c) Rs. 576.00 (d) Rs. 1152.00
44. The number of selecting three letters out of a, b, c, d, e , such that a and b are not together is
 (a) 16 (b) 18 (c) 20 (d) 14.
45. The solutions of $2x^2 + |x| - 1 = 0$ are
 (a) $x = -1, x = \frac{1}{2}$ (b) $x = 1, x = -\frac{1}{2}$
 (c) $x = \pm 1$ (d) $x = \pm \frac{1}{2}$.
46. The smallest positive number to be added to 7^3 , so that the result is divisible both by 8 and 12, is
 (a) 7 (b) 41 (c) 55 (d) 17.
47. The smallest natural number that leaves remainders 7, 8, 9 when divided by 8, 9, 10 respectively is
 (a) 359 (b) 719 (c) 361 (d) 367.

48. A mixture of volume 100 c.c. consists of two-thirds of milk and one-third water. How much water should be added to the mixture so as to make the resulting mixture contain two-thirds of water and one third of milk ?
(a) $33\frac{1}{3}$ c.c. (b) $66\frac{2}{3}$ c.c. (c) 100 c.c. (d) 50 c.c.
49. If $a+b\sqrt{2} = (\sqrt{2}+1)^5$, then a^2-2b^2 is equal to
(a) 1 (b) -1 (c) $(\sqrt{2}-1)^5$ (d) $(1-\sqrt{2})^5$.
50. The sum of products of 1, 2, 3, 4, 5 taken two at a time is
(a) 85 (b) 45 (c) 15 (d) 120.

at

ders